Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

NB SG 01

CNS Structures

Small Group Manual

Neuroscience Faculty, SGU

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Learning Objectives

A student should be able to:

SOM.MK.I.BPM2.5.NB.5.NSCI.0555 Examine the five parts of the brain and the cerebellum, including their key structures (as covered in the associated lectures) on the anatomical model of a midsagittal section of the head, while relating them to the ventricular system

SOM.MK.I.BPM2.5.NB.5.NSCI.0556 Examine the five lobes of the cerebral cortex, the boundaries separating them, and their key structures (as covered in the associated lectures) on the anatomical model of half of the brain, while relating them to the ventricular system

SOM.MK.I.BPM2.5.NB.5.NSCI.0557 Examine the diencephalon and the three parts of the brainstem, the boundaries separating them, and some of their key structures (as covered in the associated lectures) on the anatomical model of the brainstem, while relating them to the ventricular system

SOM.MK.I.BPM2.5.NB.5.NSCI.0558 Examine the elements of the ventricular system (as covered in the associated lectures) on the anatomical model of the ventricular system while relating them to the relevant structures of the central nervous system models

SOM.MK.I.BPM2.5.NB.5.NSCI.0559 Relate the functional deficits experienced by a patient in a clinical scenario to the functional anatomy of the central nervous system

What to bring to the Small Group

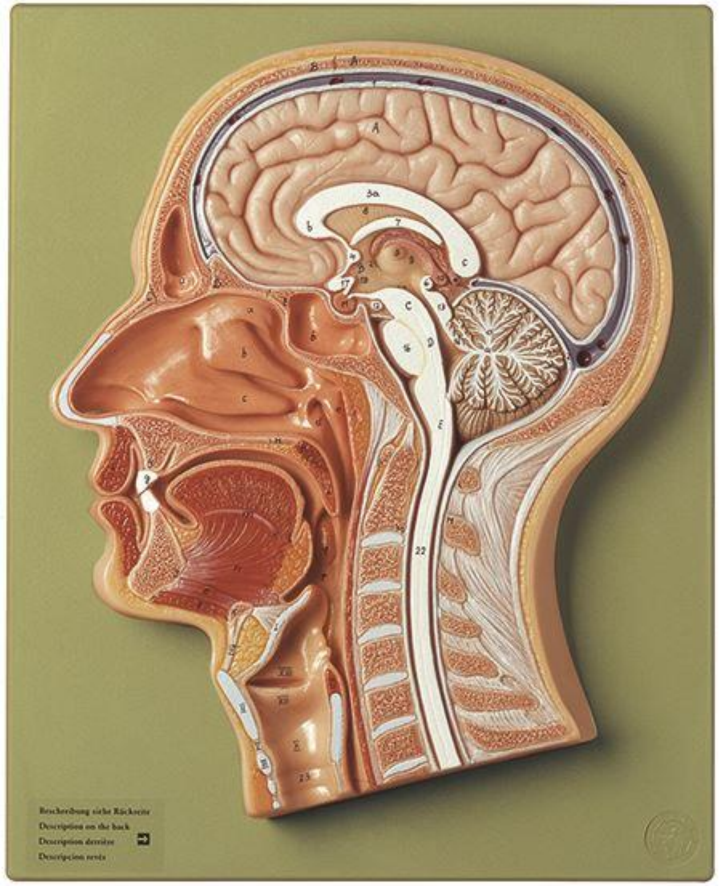
- Electronic devices, such as laptops or tablet computers, with access to small group manuals, lecture handouts, atlas and textbook

What are the Expectations

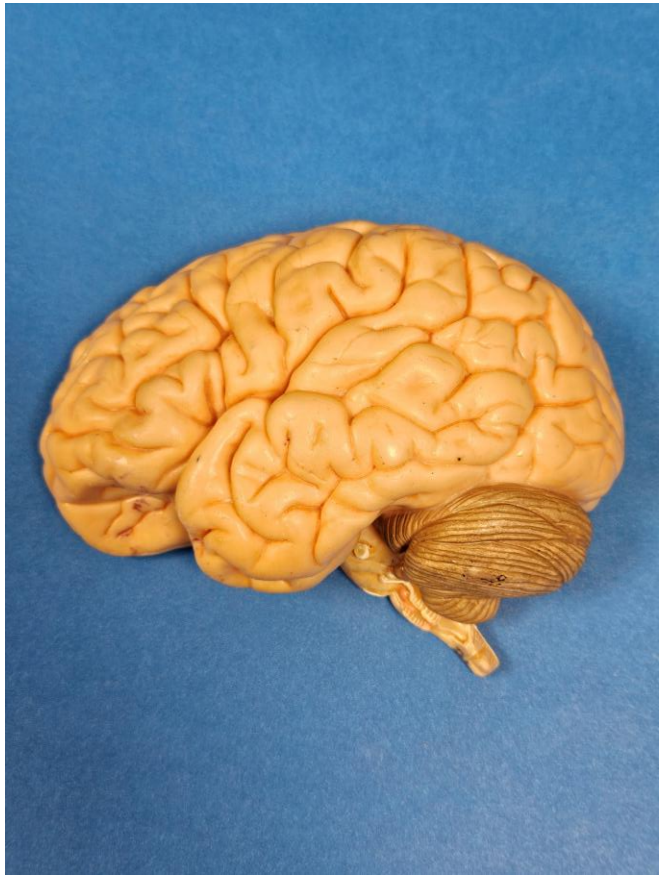
- Before the small group session
  - Review NB lectures 2, 3, and 4.
- During the small group session *)
  - Challenge yourself to work together with your peers through the material
  - Listen to your peers, ask questions, and collaboratively drive the discussion
  - Do not expect the facilitators to provide a lecture/review of the content
  - Do not expect the facilitators to drive the discussion without the group's input
  - Complete the facilitator evaluation at the end of the session

*) From: Learning Pathway (MyCourses BPM2 > Resources > General Information > Your Learning Pathway)

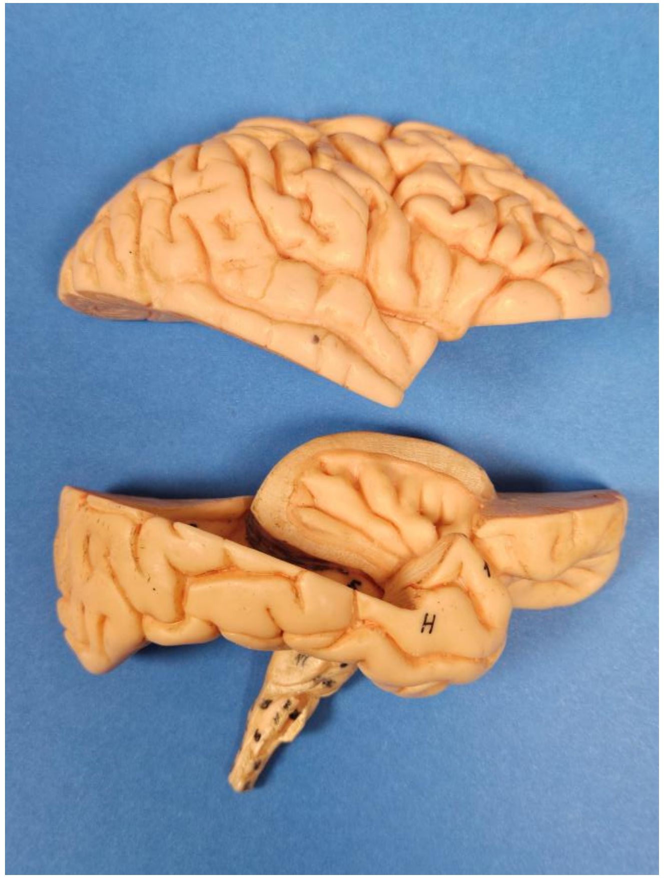
Anatomical Models provided in Small Group (1 of 2)



SOMSO Midsagittal Section of Head



SOMSO Half Brain
(or equivalent)



SOMSO Half Brain in Parts
(where available)

Anatomical Models provided in Small Group (2 of 2)

SOMSO Ventricular System

Please note: ONLY use the wooden skewers provided to point at structures of the anatomical models!

Introduction to Neurological Cases and Clinical Vignettes

- Age and sex of a patient (Example: A 44-year-old woman)
- Comes to the physician because of … (Example: a sudden onset of double vision)
- History, family history, medication
- Physical and neurologic examination
- Imaging and laboratory findings

Common Neurologic Diseases and their Characteristics (1 of 2)

Disease Category	Examples	Onset	Time Course
Vascular	Ischemic stroke (loss of blood supply) Hemorrhagic stroke (bleeding)	Typically, rapid onset	Acute (ischemic = worst at onset; hemorrhagic = getting worse)
Degenerative	Alzheimer’s disease, Parkinson’s disease	Slow onset	Chronic (gradually getting worse)
Traumatic	Injuries	Rapid onset	Acute
Infectious-inflammatory	Meningitis, Multiple sclerosis, Guillain-Barré syndrome	Variable	Acute or chronic

Common Neurologic Diseases and their Characteristics (2 of 2)

Disease Category	Examples	Onset	Time Course
Neoplastic	Tumors (glioblastomas, meningiomas)	Slow onset	Chronic (gradually getting worse)
Toxic-metabolic-nutritional	Vitamin deficiency, Poisoning	Variable	Variable
Genetic-hereditary	Huntington’s disease	May be present at birth, or manifest later in life	Chronic

Note: The disease categories are not exclusive, there are overlaps. For example, Alzheimer’s disease is degenerative, but may have a genetic component; Parkinson’s disease is degenerative and may be caused by environmental toxins.

Signs and Symptoms - Terminology

- **Symptoms** describe the **subjective** experience of a disease, condition or injury by the patient (example: headache), and usually described by the patient in a clinical setting
- **Signs** describe the **objective** indication of a disease, condition or injury, usually observed or measured by a physician or another healthcare professional, using physical examination (increased blood pressure), neurologic examination (absence of deep tendon reflexes), lab results (elevated blood sugar), or imaging (aneurysm shown in an angiogram)
- **Cardinal signs** are signs specific to a certain disease (example: resting tremor, rigidity and slowness of movement are cardinal signs of Parkinson’s disease)

Sequence of Small Group Activities

1. Students split into two subgroups of approximately 4 students each
2. Individual students present as many structures as they can identify to their subgroup, including the functions of certain structures:
   - One group starting with Midsagittal Section of the Head Model and Ventricular System
   - The other group starting with Half Brain and Half Brain in Parts models, covering superior, medial, lateral and inferior surface views
   - Then switch the models between the two groups
3. The Clinical Instructor presents the Clinical Case on the screen
4. In an interactive session, students answer the case-related questions under the guidance of the Clinical Instructor

Please note: The role of the Clinical Instructor is to guide students through the material by asking questions, or breaking down the problems into smaller components, but the focus is on student activity and participation. Come prepared, and don’t expect another lecture by the Instructor.

Questions related to the Clinical Case

1. How is the condition of the patient assessed in this video?
2. What are the signs and symptoms of the disease?
3. Based on the onset and time course of the disease process, what is the most likely category of the disease?
4. Which structure of the nervous system is most likely impacted?
5. Point out the impacted structure(s) on the models provided.
6. Which structures related to the functions assessed are most likely still intact?
7. In case the lesion would have included additional structures in the vicinity, which functions would have been most likely impacted as well?